

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 420

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 35

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 3462052667.0877614

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 187170.75660354065

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 152384.5552547754

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.96637671934

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.499999997163841

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.1330971950755286

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition -160188.41921203668

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 0.0149

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 1.1843

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 1.1611

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999663016400.225

Parameter bounds [1.000E+000; 1.000E+013]

Activation energy for condensed phase reactions 185264.1595376668

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 29589710400.44427

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 142558.41362812172

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 4.466925045124313E-05

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 1.228542096955814E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 2.136632135452227E-11

Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 274214532.11305004

Parameter bounds [1.000E+000; 1.000E+013]

Activation energy for condensed phase reactions 106661.55999422757

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 7930080.710582054

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 50064.22873680223

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.9031609962541636

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 4.119210254053302E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.4969151159245213E-07

Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 3675094668583.8755

Parameter bounds [1.000E+000; 1.000E+013]

Activation energy for condensed phase reactions 162057.39796439564

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 186095869.73846608

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 72033.34616245971

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.0038555063709016

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 2.432457327669727E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 8.881600778569264E-08

Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0237

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 15.75 mm/s

First Conditional Fuel

Surface temperature 749.28 K

Linear burning rate 11.17 mm/s

Mass decomposition rate 18.54 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 9.58e+06 W/m²

Kinetic flame height 30.88 μm
Average kinetic flame density 1.78 kg/m^3
Average kinetic flame temperature 2,228.14 K
Binder sublimation heat flux density 9.58e+06 W/m^2
Surface heat balance error -0.0013 W/m^2

Secondary Conditional Fuel

Surface temperature 741.61 K
Linear burning rate 8.54 mm/s
Mass decomposition rate 14.16 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 105,296.19 W/m^2
Kinetic flame height 1,541.73 μm
Average kinetic flame density 2.56 kg/m^3
Average kinetic flame temperature 1,553.31 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.27e+06 W/m^2
Kinetic flame height 37.46 μm
Average kinetic flame density 3.4 kg/m^3
Average kinetic flame temperature 1,166.31 K
Heat flux density due to metal combustion, 3.58e+06 W/m^2
Effective thermal conductivity 0.44 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 0.036 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.85e-05 m
Pore diameter 2.24e-05 m
Diffusion flame heat flux density 17,946.99 W/m^2
Diffusion flame height 15,949.11 μm
Heat flux density to surface within skeletal structure 2.1e+06 W/m^2
Heat flux density to surface outside skeletal structure 67,501.96 W/m^2
Total heat flux density to surface 7.16e+06 W/m^2
Binder sublimation heat flux density 7.16e+06 W/m^2
Surface heat balance error -2.56e-04 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 35.34 mm/s

First Conditional Fuel

Surface temperature 802.14 K
Linear burning rate 62.04 mm/s
Mass decomposition rate 102.92 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 6.13e+07 W/m^2
Kinetic flame height 4.74 μm
Average kinetic flame density 7.14 kg/m^3
Average kinetic flame temperature 2,254.57 K
Binder sublimation heat flux density 6.13e+07 W/m^2
Surface heat balance error 0.012 W/m^2

Secondary Conditional Fuel

Surface temperature 759.82 K
Linear burning rate 16.03 mm/s
Mass decomposition rate 26.59 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 275,373.96 W/m^2
Kinetic flame height 582.91 μm
Average kinetic flame density 10.3 kg/m^3
Average kinetic flame temperature 1,562.41 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.07e+06 W/m^2
Kinetic flame height 16.41 μm

Average kinetic flame density 13.69 kg/m³
Average kinetic flame temperature 1,175.41 K
Heat flux density due to metal combustion, 7e+06 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.017 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.25e-05 m
Pore diameter 1.08e-05 m
Diffusion flame heat flux density 154,968.01 W/m²
Diffusion flame height 1,835.33 μm
Heat flux density to surface within skeletal structure 3.98e+06 W/m²
Heat flux density to surface outside skeletal structure 184,405.2 W/m²
Total heat flux density to surface 1.42e+07 W/m²
Binder sublimation heat flux density 1.42e+07 W/m²
Surface heat balance error -5.25e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 40.4 mm/s

First Conditional Fuel

Surface temperature 821.7 K
Linear burning rate 110.62 mm/s
Mass decomposition rate 183.53 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.15e+08 W/m²
Kinetic flame height 2.51 μm
Average kinetic flame density 11.39 kg/m³
Average kinetic flame temperature 2,264.35 K
Binder sublimation heat flux density 1.15e+08 W/m²
Surface heat balance error -0.017 W/m²

Secondary Conditional Fuel

Surface temperature 762.65 K
Linear burning rate 17.63 mm/s
Mass decomposition rate 29.24 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 427,624.49 W/m²
Kinetic flame height 374.71 μm
Average kinetic flame density 16.5 kg/m³
Average kinetic flame temperature 1,563.82 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 7.43e+06 W/m²
Kinetic flame height 11.15 μm
Average kinetic flame density 21.92 kg/m³
Average kinetic flame temperature 1,176.82 K
Heat flux density due to metal combustion, 7.75e+06 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.016 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.9e-05 m
Pore diameter 9.66e-06 m
Diffusion flame heat flux density 356,715.91 W/m²
Diffusion flame height 796.53 μm
Heat flux density to surface within skeletal structure 4.12e+06 W/m²
Heat flux density to surface outside skeletal structure 311,447.58 W/m²
Total heat flux density to surface 1.57e+07 W/m²
Binder sublimation heat flux density 1.57e+07 W/m²
Surface heat balance error -0.0012 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 14.92 mm/s

First Conditional Fuel

Surface temperature 749.28 K
Linear burning rate 11.17 mm/s
Mass decomposition rate $18.54 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.58\text{e}+06 \text{ W/m}^2$
Kinetic flame height 30.88 μm
Average kinetic flame density 1.78 kg/m^3
Average kinetic flame temperature 2,228.14 K
Binder sublimation heat flux density $9.58\text{e}+06 \text{ W/m}^2$
Surface heat balance error -0.0013 W/m^2

Secondary Conditional Fuel

Surface temperature 749.28 K
Linear burning rate 11.17 mm/s
Mass decomposition rate $18.54 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $80,620.33 \text{ W/m}^2$
Kinetic flame height 2,004.1 μm
Average kinetic flame density 2.55 kg/m^3
Average kinetic flame temperature 1,557.14 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.61\text{e}+07 \text{ W/m}^2$
Kinetic flame height 7.19 μm
Average kinetic flame density 2.35 kg/m^3
Average kinetic flame temperature 1,686.14 K
Heat flux density due to metal combustion, $4.25\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.38 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.16 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E-}08$
Skeleton layer thickness $4.98\text{e-}05 \text{ m}$
Pore diameter $1.64\text{e-}05 \text{ m}$
Diffusion flame heat flux density $37,050.73 \text{ W/m}^2$
Diffusion flame height 7,893.81 μm
Heat flux density to surface within skeletal structure $9.49\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $55,393.38 \text{ W/m}^2$
Total heat flux density to surface $9.58\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $9.58\text{e}+06 \text{ W/m}^2$
Surface heat balance error 0.0011 W/m^2

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 35.87 mm/s

First Conditional Fuel

Surface temperature 802.14 K
Linear burning rate 62.04 mm/s
Mass decomposition rate $102.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $6.13\text{e}+07 \text{ W/m}^2$
Kinetic flame height 4.74 μm
Average kinetic flame density 7.14 kg/m^3
Average kinetic flame temperature 2,254.57 K
Binder sublimation heat flux density $6.13\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.012 W/m^2

Secondary Conditional Fuel

Surface temperature 764.61 K

Linear burning rate 18.82 mm/s

Mass decomposition rate $31.22 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 234,782.44 W/m²

Kinetic flame height 681.65 μm

Average kinetic flame density 10.29 kg/m^3

Average kinetic flame temperature 1,564.8 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.38\text{e}+07 \text{ W/m}^2$

Kinetic flame height 2.91 μm

Average kinetic flame density 9.5 kg/m^3

Average kinetic flame temperature 1,693.8 K

Heat flux density due to metal combustion, $6.18\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.31 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.089 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness $2.69\text{e}-05 \text{ m}$

Pore diameter $8.95\text{e}-06 \text{ m}$

Diffusion flame heat flux density 356,855.5 W/m²

Diffusion flame height 815.29 μm

Heat flux density to surface within skeletal structure $1.63\text{e}+07 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure 180,035.82 W/m²

Total heat flux density to surface $1.69\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $1.69\text{e}+07 \text{ W/m}^2$

Surface heat balance error $-9.75\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s

Calculated burning rate 39.91 mm/s

First Conditional Fuel

Surface temperature 821.7 K

Linear burning rate 110.62 mm/s

Mass decomposition rate $183.53 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $1.15\text{e}+08 \text{ W/m}^2$

Kinetic flame height 2.51 μm

Average kinetic flame density 11.39 kg/m^3

Average kinetic flame temperature 2,264.35 K

Binder sublimation heat flux density $1.15\text{e}+08 \text{ W/m}^2$

Surface heat balance error -0.017 W/m^2

Secondary Conditional Fuel

Surface temperature 765.34 K

Linear burning rate 19.28 mm/s

Mass decomposition rate $31.99 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 391,126.12 W/m²

Kinetic flame height 408.99 μm

Average kinetic flame density 16.48 kg/m^3

Average kinetic flame temperature 1,565.17 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.86 μm

Average kinetic flame density 15.23 kg/m^3

Average kinetic flame temperature 1,694.17 K

Heat flux density due to metal combustion, $6.3\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.31 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.086 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 2.61×10^{-5} m
Pore diameter 8.7×10^{-6} m
Diffusion flame heat flux density $881,930.18 \text{ W/m}^2$
Diffusion flame height $329.81 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $1.61 \times 10^7 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $331,938.17 \text{ W/m}^2$
Total heat flux density to surface $1.73 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $1.73 \times 10^7 \text{ W/m}^2$
Surface heat balance error -0.0048 W/m^2

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 10.79 mm/s

First Conditional Fuel

Surface temperature 749.28 K
Linear burning rate 11.17 mm/s
Mass decomposition rate $18.54 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.58 \times 10^6 \text{ W/m}^2$
Kinetic flame height $30.88 \text{ }\mu\text{m}$
Average kinetic flame density 1.78 kg/m^3
Average kinetic flame temperature $2,228.14 \text{ K}$
Binder sublimation heat flux density $9.58 \times 10^6 \text{ W/m}^2$
Surface heat balance error -0.0013 W/m^2

Secondary Conditional Fuel

Surface temperature 719.3 K
Linear burning rate 3.78 mm/s
Mass decomposition rate $6.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $4,676.53 \text{ W/m}^2$
Kinetic flame height $6,900.5 \text{ }\mu\text{m}$
Average kinetic flame density 4.51 kg/m^3
Average kinetic flame temperature 880.65 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $914,293.46 \text{ W/m}^2$
Kinetic flame height $57.06 \text{ }\mu\text{m}$
Average kinetic flame density 4.05 kg/m^3
Average kinetic flame temperature 980.15 K
Heat flux density due to metal combustion, $4.66 \times 10^6 \text{ W/m}^2$
Effective thermal conductivity $1.45 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.035 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641 \times 10^{-10}$
Skeleton layer thickness $1.8 \times 10^{-4} \text{ m}$
Pore diameter $5.77 \times 10^{-5} \text{ m}$
Diffusion flame heat flux density $4,697.97 \text{ W/m}^2$
Diffusion flame height $60,892.34 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $2.95 \times 10^6 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $2,203.73 \text{ W/m}^2$
Total heat flux density to surface $2.96 \times 10^6 \text{ W/m}^2$
Binder sublimation heat flux density $2.96 \times 10^6 \text{ W/m}^2$
Surface heat balance error $-5.06 \times 10^{-5} \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 26.22 mm/s

First Conditional Fuel

Surface temperature 802.14 K

Linear burning rate 62.04 mm/s
Mass decomposition rate $102.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $6.13\text{e}+07 \text{ W/m}^2$
Kinetic flame height 4.74 μm
Average kinetic flame density 7.14 kg/m^3
Average kinetic flame temperature 2,254.57 K
Binder sublimation heat flux density $6.13\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.012 W/m^2

Secondary Conditional Fuel

Surface temperature 739.74 K
Linear burning rate 7.99 mm/s
Mass decomposition rate $13.25 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $10,802.45 \text{ W/m}^2$
Kinetic flame height 2,798.02 μm
Average kinetic flame density 18.07 kg/m^3
Average kinetic flame temperature 890.87 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.84\text{e}+06 \text{ W/m}^2$
Kinetic flame height 27.3 μm
Average kinetic flame density 16.25 kg/m^3
Average kinetic flame temperature 990.37 K
Heat flux density due to metal combustion, $1.08\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.43 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.015 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $7.41\text{e}-05 \text{ m}$
Pore diameter $2.42\text{e}-05 \text{ m}$

Diffusion flame heat flux density $36,097.7 \text{ W/m}^2$
Diffusion flame height 7,868.24 μm
Heat flux density to surface within skeletal structure $6.62\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $5,131.38 \text{ W/m}^2$
Total heat flux density to surface $6.66\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $6.66\text{e}+06 \text{ W/m}^2$
Surface heat balance error $2.47\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 34.58 mm/s

First Conditional Fuel

Surface temperature 821.7 K
Linear burning rate 110.62 mm/s
Mass decomposition rate $183.53 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.15\text{e}+08 \text{ W/m}^2$
Kinetic flame height 2.51 μm
Average kinetic flame density 11.39 kg/m^3
Average kinetic flame temperature 2,264.35 K
Binder sublimation heat flux density $1.15\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.017 W/m^2

Secondary Conditional Fuel

Surface temperature 746.85 K
Linear burning rate 10.26 mm/s
Mass decomposition rate $17.03 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $14,327.02 \text{ W/m}^2$
Kinetic flame height 2,060.1 μm
Average kinetic flame density 28.84 kg/m^3
Average kinetic flame temperature 894.42 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.33\text{e}+06 \text{ W/m}^2$
Kinetic flame height $21.25 \mu\text{m}$
Average kinetic flame density 25.96 kg/m^3
Average kinetic flame temperature 993.92 K
Heat flux density due to metal combustion, $1.43\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.011 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $5.51\text{e}-05 \text{ m}$
Pore diameter $1.81\text{e}-05 \text{ m}$
Diffusion flame heat flux density $71,411.32 \text{ W/m}^2$
Diffusion flame height $3,967.37 \mu\text{m}$
Heat flux density to surface within skeletal structure $8.66\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $6,851.72 \text{ W/m}^2$
Total heat flux density to surface $8.74\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $8.74\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-1.39\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 14.22 mm/s

First Conditional Fuel

Surface temperature 776.12 K
Linear burning rate 10.96 mm/s
Mass decomposition rate $18.18 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.01\text{e}+07 \text{ W/m}^2$
Kinetic flame height $28.94 \mu\text{m}$
Average kinetic flame density 1.77 kg/m^3
Average kinetic flame temperature $2,241.56 \text{ K}$
Binder sublimation heat flux density $1.01\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.001 W/m^2

Secondary Conditional Fuel

Surface temperature 758.79 K
Linear burning rate 7.51 mm/s
Mass decomposition rate $12.47 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $120,174.7 \text{ W/m}^2$
Kinetic flame height $1,336.56 \mu\text{m}$
Average kinetic flame density 2.54 kg/m^3
Average kinetic flame temperature $1,561.89 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $3.19\text{e}+06 \text{ W/m}^2$
Kinetic flame height $26.06 \mu\text{m}$
Average kinetic flame density 3.38 kg/m^3
Average kinetic flame temperature $1,174.89 \text{ K}$
Heat flux density due to metal combustion, $1.9\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.47 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
Pore diameter $4.38\text{e}-05 \text{ m}$
Diffusion flame heat flux density $20,712.29 \text{ W/m}^2$
Diffusion flame height $14,036.16 \mu\text{m}$
Heat flux density to surface within skeletal structure $1.93\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure 74,742.14 W/m²
Total heat flux density to surface 6.62e+06 W/m²
Binder sublimation heat flux density 6.62e+06 W/m²
Surface heat balance error -9.16e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 37.88 mm/s

First Conditional Fuel

Surface temperature 865.12 K
Linear burning rate 60.02 mm/s
Mass decomposition rate 99.57 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 6.88e+07 W/m²
Kinetic flame height 4.13 μm
Average kinetic flame density 7.04 kg/m³
Average kinetic flame temperature 2,286.06 K
Binder sublimation heat flux density 6.88e+07 W/m²
Surface heat balance error -0.0023 W/m²

Secondary Conditional Fuel

Surface temperature 798.41 K
Linear burning rate 17.39 mm/s
Mass decomposition rate 28.85 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 256,048.11 W/m²
Kinetic flame height 611.83 μm
Average kinetic flame density 10.18 kg/m³
Average kinetic flame temperature 1,581.71 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.99e+07 W/m²
Kinetic flame height 3.98 μm
Average kinetic flame density 13.47 kg/m³
Average kinetic flame temperature 1,194.71 K
Heat flux density due to metal combustion, 4.32e+06 W/m²
Effective thermal conductivity 0.43 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.027 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5e-05 m
Pore diameter 1.65e-05 m
Diffusion flame heat flux density 144,018.26 W/m²
Diffusion flame height 1,991.13 μm
Heat flux density to surface within skeletal structure 4.85e+06 W/m²
Heat flux density to surface outside skeletal structure 204,762.63 W/m²
Total heat flux density to surface 1.7e+07 W/m²
Binder sublimation heat flux density 1.7e+07 W/m²
Surface heat balance error -0.0011 W/m²

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
Calculated burning rate 52.94 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 106.63 mm/s
Mass decomposition rate 176.9 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.31e+08 W/m²
Kinetic flame height 2.14 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 1.31e+08 W/m²

Surface heat balance error 0.0012 W/m²

Secondary Conditional Fuel

Surface temperature 814.05 K

Linear burning rate 23.67 mm/s

Mass decomposition rate 39.27 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 321,929.72 W/m²

Kinetic flame height 481.77 μm

Average kinetic flame density 16.23 kg/m³

Average kinetic flame temperature 1,589.52 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.59e+07 W/m²

Kinetic flame height 2.16 μm

Average kinetic flame density 21.45 kg/m³

Average kinetic flame temperature 1,202.52 K

Heat flux density due to metal combustion, 5.92e+06 W/m²

Effective thermal conductivity 0.42 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 0.019 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 3.47e-05 m

Pore diameter 1.16e-05 m

Diffusion flame heat flux density 266,573.42 W/m²

Diffusion flame height 1,069.86 μm

Heat flux density to surface within skeletal structure 6.83e+06 W/m²

Heat flux density to surface outside skeletal structure 269,370.06 W/m²

Total heat flux density to surface 2.41e+07 W/m²

Binder sublimation heat flux density 2.41e+07 W/m²

Surface heat balance error 3.43e-06 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s

Calculated burning rate 5.97 mm/s

First Conditional Fuel

Surface temperature 823.38 K

Linear burning rate 10.65 mm/s

Mass decomposition rate 17.67 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.11e+07 W/m²

Kinetic flame height 26 μm

Average kinetic flame density 1.75 kg/m³

Average kinetic flame temperature 2,265.19 K

Binder sublimation heat flux density 1.11e+07 W/m²

Surface heat balance error -6.29e-04 W/m²

Secondary Conditional Fuel

Surface temperature 783.69 K

Linear burning rate 2.71 mm/s

Mass decomposition rate 4.49 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 335,752.41 W/m²

Kinetic flame height 470.97 μm

Average kinetic flame density 2.52 kg/m³

Average kinetic flame temperature 1,574.35 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 284,941.06 W/m²

Kinetic flame height 283.32 μm

Average kinetic flame density 3.34 kg/m³

Average kinetic flame temperature 1,187.35 K

Heat flux density due to metal combustion, $1.54\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $3.29\text{e}-05 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
Pore diameter $2.05\text{e}-08 \text{ m}$
Diffusion flame heat flux density $57,038.24 \text{ W/m}^2$
Diffusion flame height $5,053.29 \mu\text{m}$
Heat flux density to surface within skeletal structure $472,862.83 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $248,929.72 \text{ W/m}^2$
Total heat flux density to surface $2.55\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $2.55\text{e}+06 \text{ W/m}^2$
Surface heat balance error $3.87\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s
Calculated burning rate 13.65 mm/s

First Conditional Fuel

Surface temperature 879.36 K
Linear burning rate 59.64 mm/s
Mass decomposition rate $98.94 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $7.04\text{e}+07 \text{ W/m}^2$
Kinetic flame height $4.01 \mu\text{m}$
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature $2,293.18 \text{ K}$
Binder sublimation heat flux density $7.04\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.012 W/m^2

Secondary Conditional Fuel

Surface temperature 805.33 K
Linear burning rate 5.81 mm/s
Mass decomposition rate $9.63 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $767,864.35 \text{ W/m}^2$
Kinetic flame height $203.12 \mu\text{m}$
Average kinetic flame density 10.15 kg/m^3
Average kinetic flame temperature $1,585.16 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $147,203.02 \text{ W/m}^2$
Kinetic flame height $533.73 \mu\text{m}$
Average kinetic flame density 13.43 kg/m^3
Average kinetic flame temperature $1,198.16 \text{ K}$
Heat flux density due to metal combustion, $3.65\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.36\text{e}-05 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $5.46\text{e}-05 \text{ m}$
Pore diameter $8.43\text{e}-09 \text{ m}$
Diffusion flame heat flux density $430,254.97 \text{ W/m}^2$
Diffusion flame height $664.88 \mu\text{m}$
Heat flux density to surface within skeletal structure $712,909.05 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $623,844.35 \text{ W/m}^2$
Total heat flux density to surface $5.79\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $5.79\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-1.55\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 18 mm/s

Calculated burning rate 17.99 mm/s

First Conditional Fuel

Surface temperature 900 K

Linear burning rate 106.63 mm/s

Mass decomposition rate $176.9 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Kinetic flame heat flux density to surface $1.31\text{e}+08 \text{ W/m}^2$

Kinetic flame height 2.14 μm

Average kinetic flame density 11.2 kg/m^3

Average kinetic flame temperature 2,303.5 K

Binder sublimation heat flux density $1.31\text{e}+08 \text{ W/m}^2$

Surface heat balance error -0.026 W/m^2

Secondary Conditional Fuel

Surface temperature 813.13 K

Linear burning rate 7.57 mm/s

Mass decomposition rate $12.56 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.01\text{e}+06 \text{ W/m}^2$

Kinetic flame height 154.19 μm

Average kinetic flame density 16.23 kg/m^3

Average kinetic flame temperature 1,589.06 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $117,080.66 \text{ W/m}^2$

Kinetic flame height 664.39 μm

Average kinetic flame density 21.46 kg/m^3

Average kinetic flame temperature 1,202.06 K

Heat flux density due to metal combustion, $4.92\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Radiative thermal conductivity $9.97\text{e}-06 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$

Skeleton layer thickness $3.99\text{e}-05 \text{ m}$

Pore diameter $6.2\text{e}-09 \text{ m}$

Diffusion flame heat flux density $833,817.76 \text{ W/m}^2$

Diffusion flame height 342.15 μm

Heat flux density to surface within skeletal structure $635,631.63 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $879,569.88 \text{ W/m}^2$

Total heat flux density to surface $7.69\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $7.69\text{e}+06 \text{ W/m}^2$

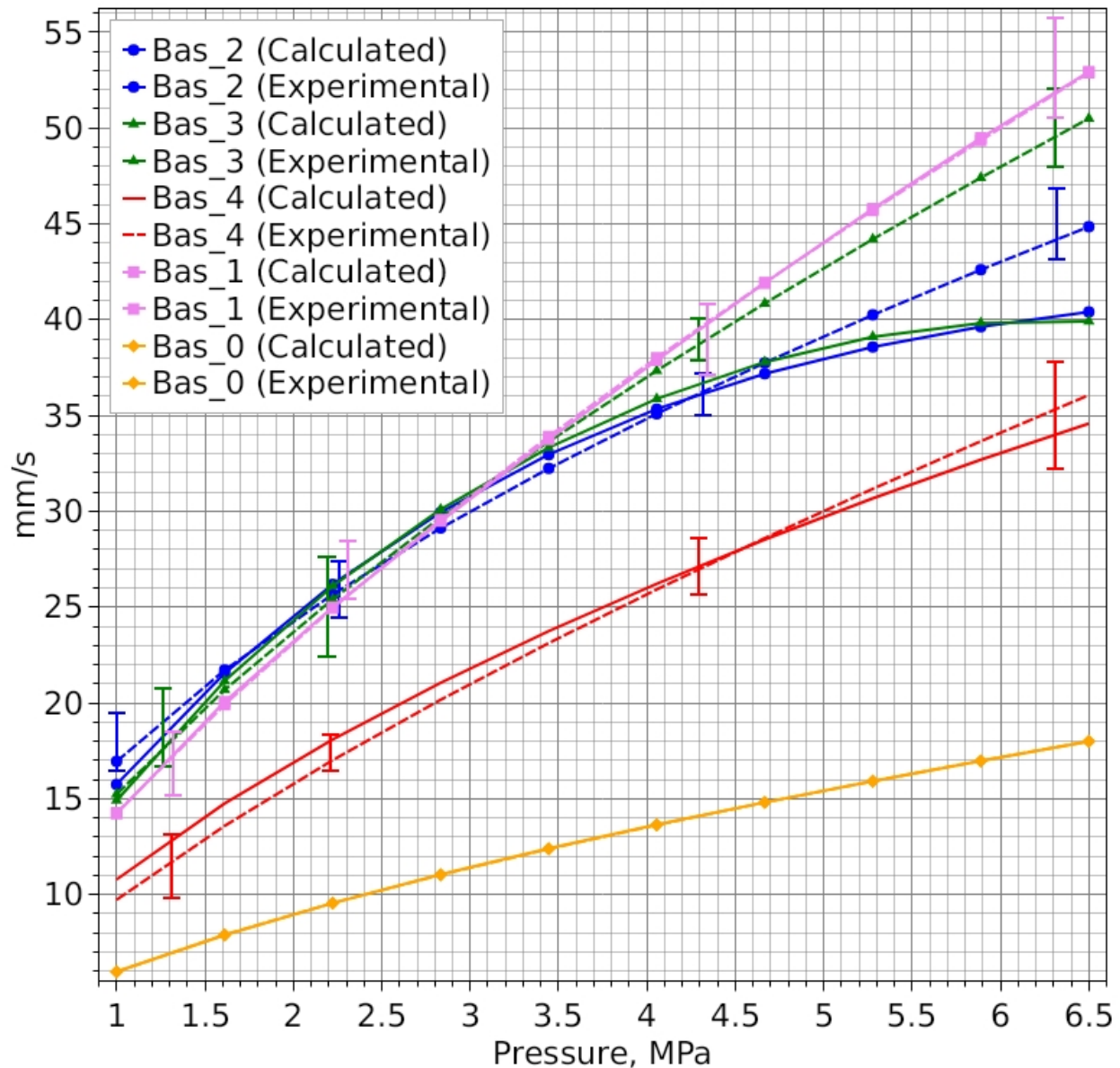
Surface heat balance error $-5.83\text{e}-04 \text{ W/m}^2$

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	15.75 mm/s		14.92 mm/s		10.79 mm/s		14.22 mm/s		5.97 mm/s	
Difference	1.19 mm/s		0.32 mm/s		-1.06 mm/s		0.042 mm/s		-0.0084 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.17 mm/s	8.54 mm/s	11.17 mm/s	11.17 mm/s	11.17 mm/s	3.78 mm/s	10.96 mm/s	7.51 mm/s	10.65 mm/s	2.71 mm/s
Surface Temperature	749.28 K	741.61 K	749.28 K	749.28 K	749.28 K	719.3 K	776.12 K	758.79 K	823.38 K	783.69 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	9.58e+06 W/m ²		9.58e+06 W/m ²		9.58e+06 W/m ²		1.01e+07 W/m ²		1.11e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	2.27e+06 W/m ²		2.61e+07 W/m ²		914,293.46 W/m ²		3.19e+06 W/m ²		284,941.06 W/m ²	
Kinetic Flame Height (Skeleton)	37.46 μm		7.19 μm		57.06 μm		26.06 μm		283.32 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	105,296.19 W/m ²		80,620.33 W/m ²		4,676.53 W/m ²		120,174.7 W/m ²		335,752.41 W/m ²	
Kinetic Flame Height (Out-Skeleton)	1,541.73 μm		2,004.1 μm		6,900.5 μm		1,336.56 μm		470.97 μm	
Diffusion Flame Heat Flux Density	17,946.99 W/m ²		37,050.73 W/m ²		4,697.97 W/m ²		20,712.29 W/m ²		57,038.24 W/m ²	
Diffusion Flame Height	15,949.11 μm		7,893.81 μm		60,892.34 μm		14,036.16 μm		5,053.29 μm	
Metal Combustion Heat Flux Density	3.58e+06 W/m ²		4.25e+06 W/m ²		4.66e+06 W/m ²		1.9e+06 W/m ²		1.54e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	35.34 mm/s		35.87 mm/s		26.22 mm/s		37.88 mm/s		13.65 mm/s	
Difference	-0.24 mm/s		1.47 mm/s		-0.3 mm/s		0.13 mm/s		-0.021 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	62.04 mm/s	16.03 mm/s	62.04 mm/s	18.82 mm/s	62.04 mm/s	7.99 mm/s	60.02 mm/s	17.39 mm/s	59.64 mm/s	5.81 mm/s
Surface Temperature	802.14 K	759.82 K	802.14 K	764.61 K	802.14 K	739.74 K	865.12 K	798.41 K	879.36 K	805.33 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	6.13e+07 W/m ²		6.13e+07 W/m ²		6.13e+07 W/m ²		6.88e+07 W/m ²		7.04e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	5.07e+06 W/m ²		6.38e+07 W/m ²		1.84e+06 W/m ²		1.99e+07 W/m ²		147,203.02 W/m ²	
Kinetic Flame Height (Skeleton)	16.41 μm		2.91 μm		27.3 μm		3.98 μm		533.73 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	275,373.96 W/m ²		234,782.44 W/m ²		10,802.45 W/m ²		256,048.11 W/m ²		767,864.35 W/m ²	
Kinetic Flame Height (Out-Skeleton)	582.91 μm		681.65 μm		2,798.02 μm		611.83 μm		203.12 μm	
Diffusion Flame Heat Flux Density	154,968.01 W/m ²		356,855.5 W/m ²		36,097.7 W/m ²		144,018.26 W/m ²		430,254.97 W/m ²	
Diffusion Flame Height	1,835.33 μm		815.29 μm		7,868.24 μm		1,991.13 μm		664.88 μm	
Metal Combustion Heat Flux Density	7e+06 W/m ²		6.18e+06 W/m ²		1.08e+07 W/m ²		4.32e+06 W/m ²		3.65e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	40.4 mm/s		39.91 mm/s		34.58 mm/s		52.94 mm/s		17.99 mm/s	
Difference	4.44 mm/s		10.59 mm/s		1.49 mm/s		-0.056 mm/s		0.0078 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	110.62 mm/s	17.63 mm/s	110.62 mm/s	19.28 mm/s	110.62 mm/s	10.26 mm/s	106.63 mm/s	23.67 mm/s	106.63 mm/s	7.57 mm/s
Surface Temperature	821.7 K	762.65 K	821.7 K	765.34 K	821.7 K	746.85 K	900 K	814.05 K	900 K	813.13 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.15e+08 W/m ²		1.15e+08 W/m ²		1.15e+08 W/m ²		1.31e+08 W/m ²		1.31e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	7.43e+06 W/m ²		1e+08 W/m ²		2.33e+06 W/m ²		3.59e+07 W/m ²		117,080.66 W/m ²	
Kinetic Flame Height (Skeleton)	11.15 μm		1.86 μm		21.25 μm		2.16 μm		664.39 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	427,624.49 W/m ²		391,126.12 W/m ²		14,327.02 W/m ²		321,929.72 W/m ²		1.01e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	374.71 μm		408.99 μm		2,060.1 μm		481.77 μm		154.19 μm	
Diffusion Flame Heat Flux Density	356,715.91 W/m ²		881,930.18 W/m ²		71,411.32 W/m ²		266,573.42 W/m ²		833,817.76 W/m ²	
Diffusion Flame Height	796.53 μm		329.81 μm		3,967.37 μm		1,069.86 μm		342.15 μm	
Metal Combustion Heat Flux Density	7.75e+06 W/m ²		6.3e+06 W/m ²		1.43e+07 W/m ²		5.92e+06 W/m ²		4.92e+06 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341\text{E-}05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

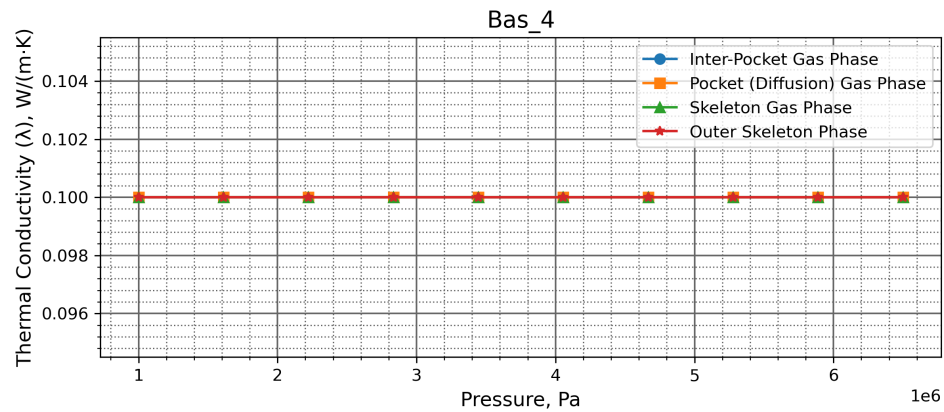
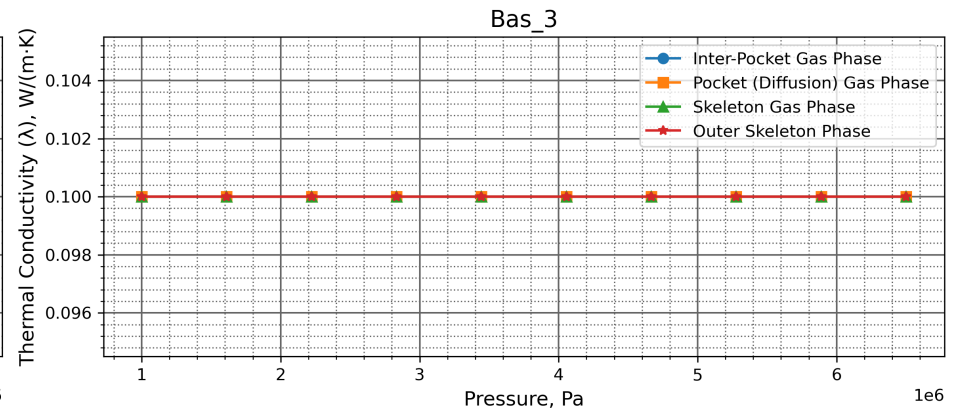
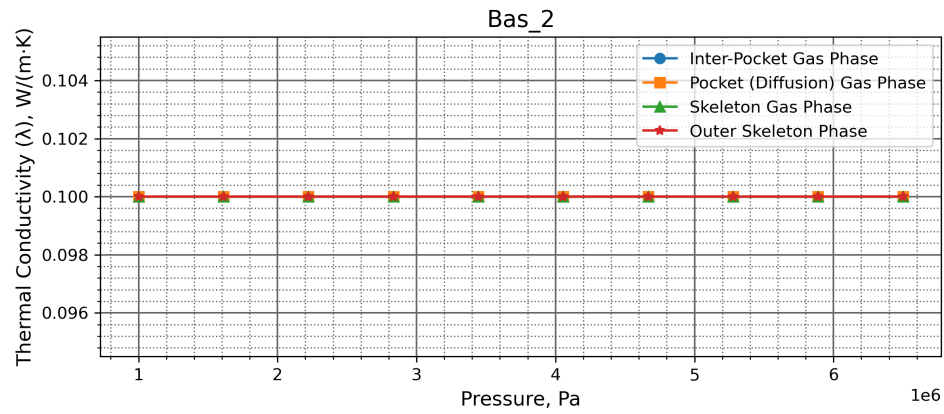
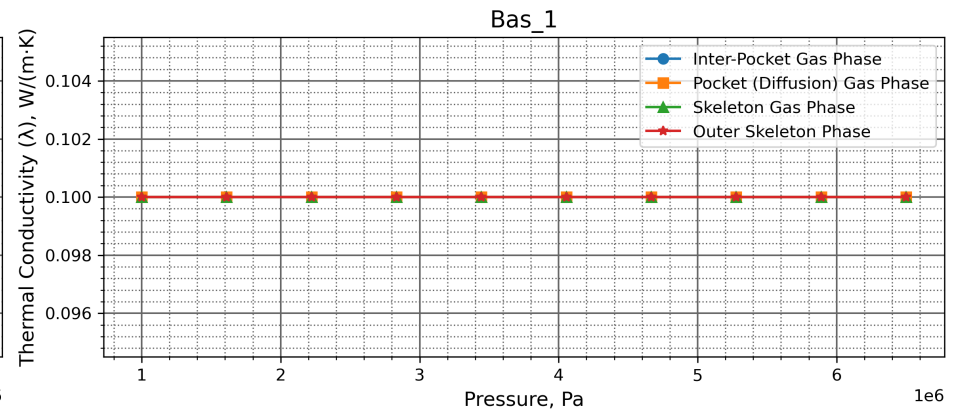
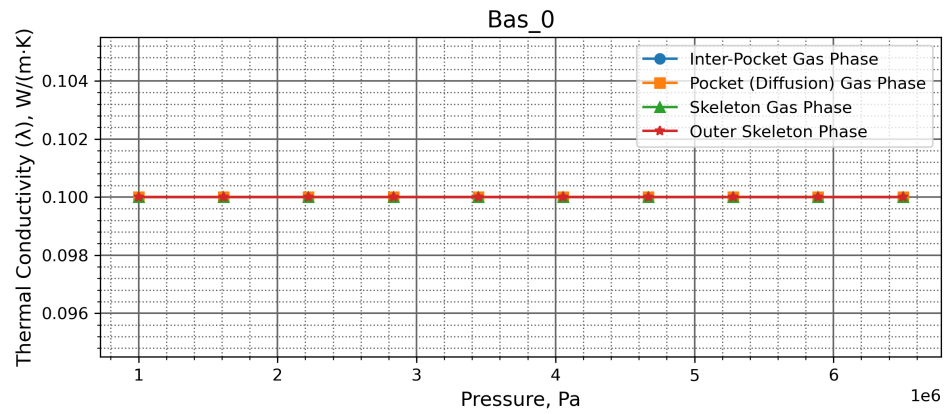
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

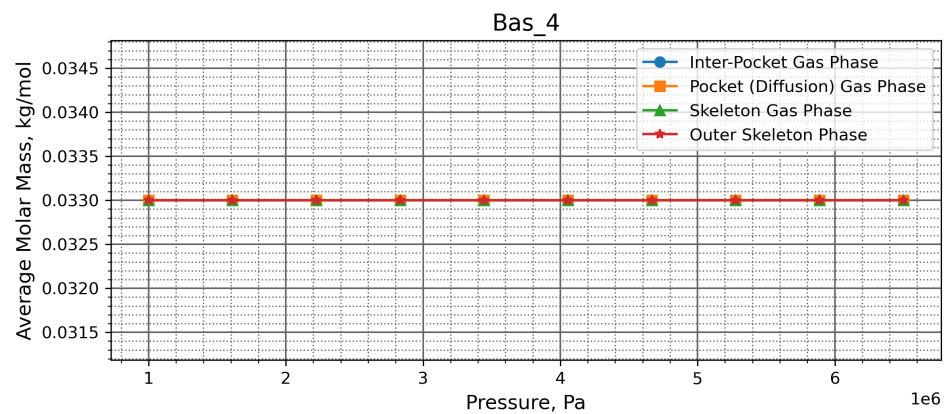
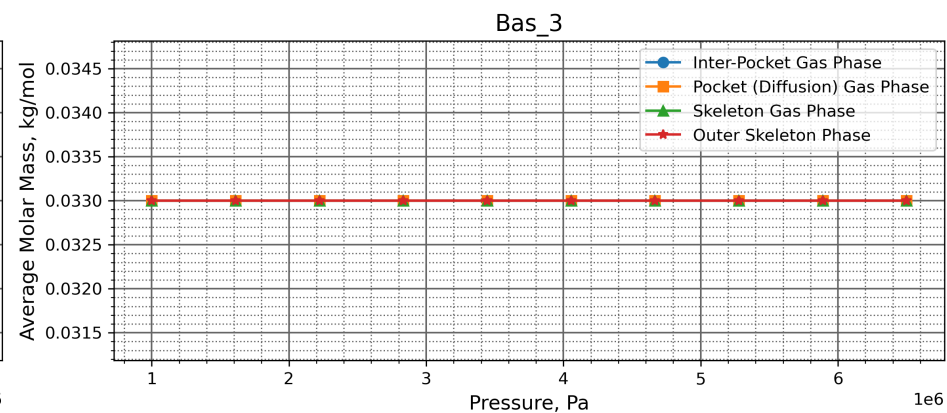
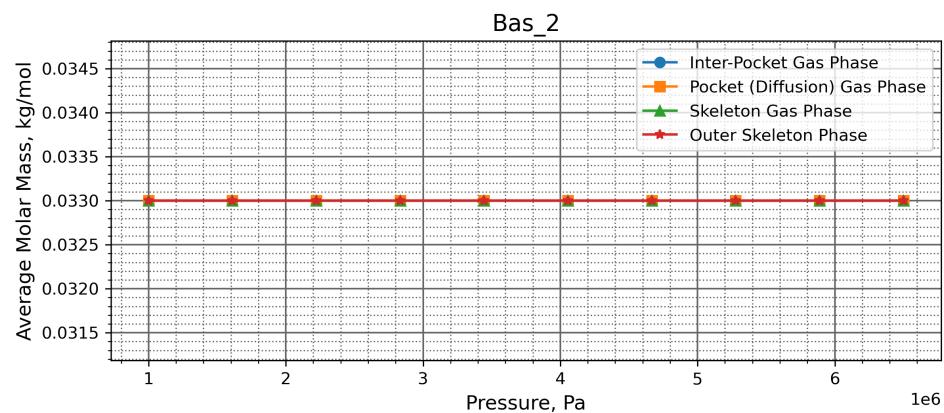
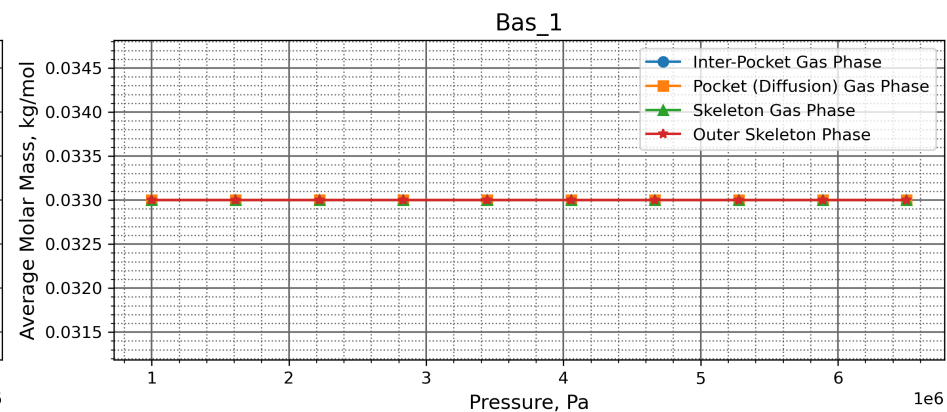
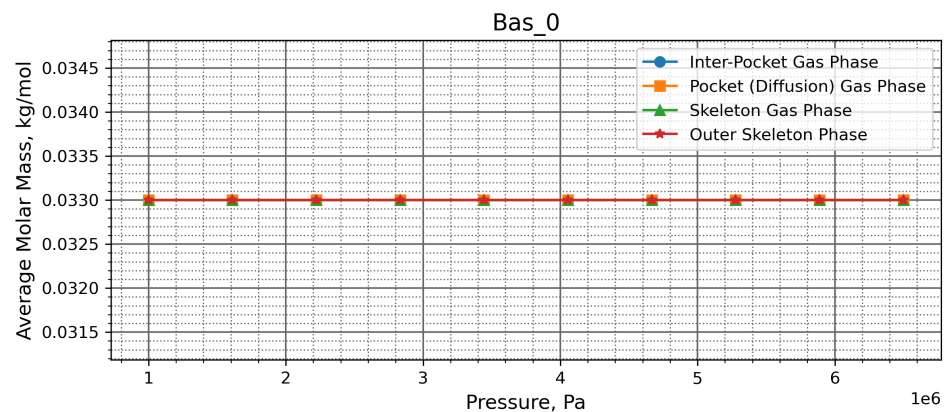
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

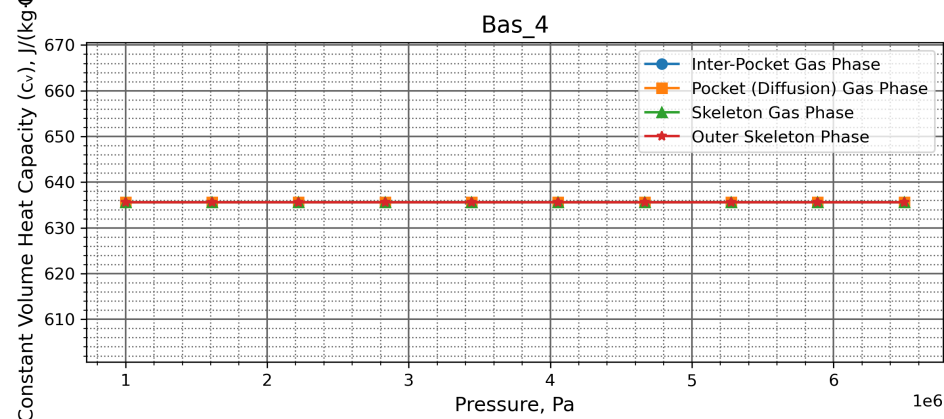
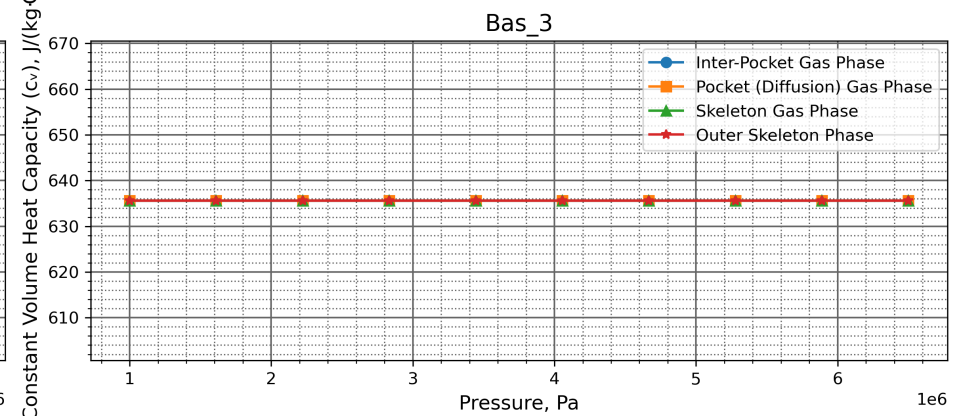
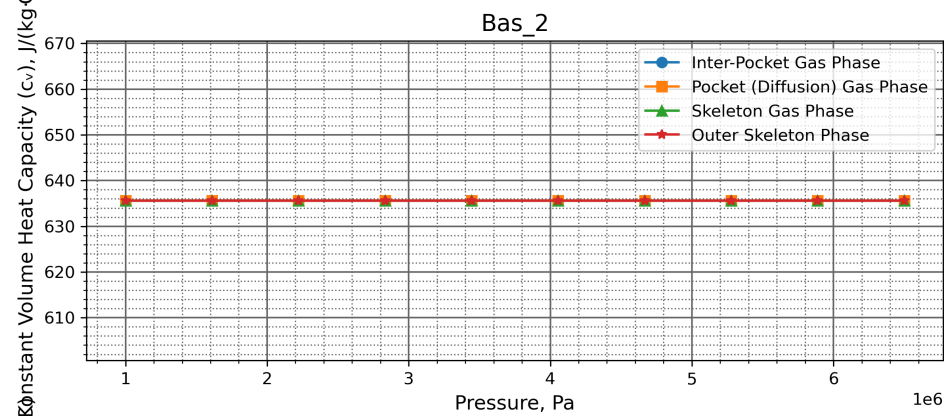
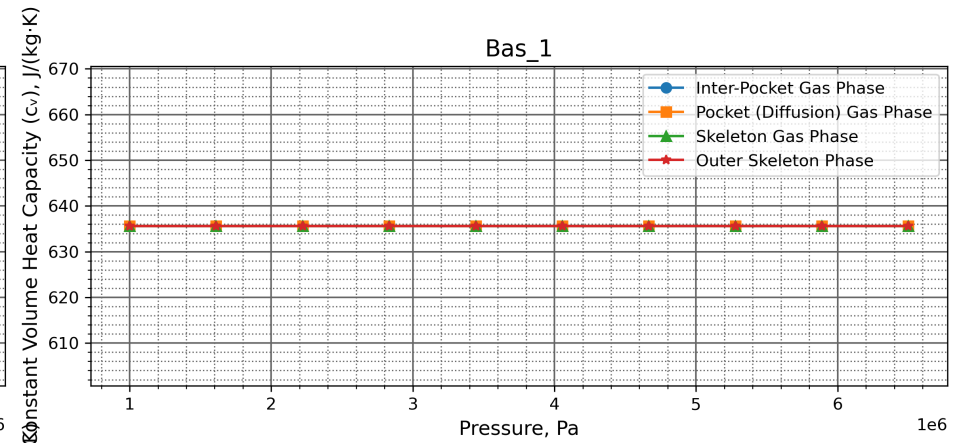
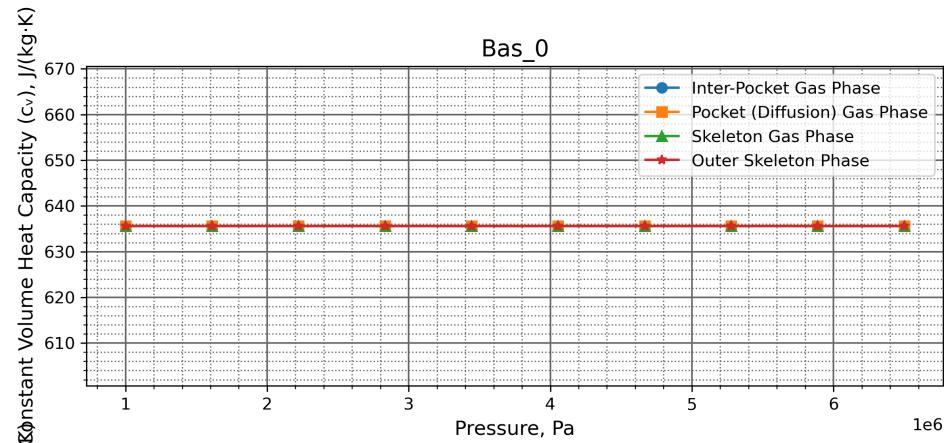
Thermal Conductivity (λ), W/(m·K)



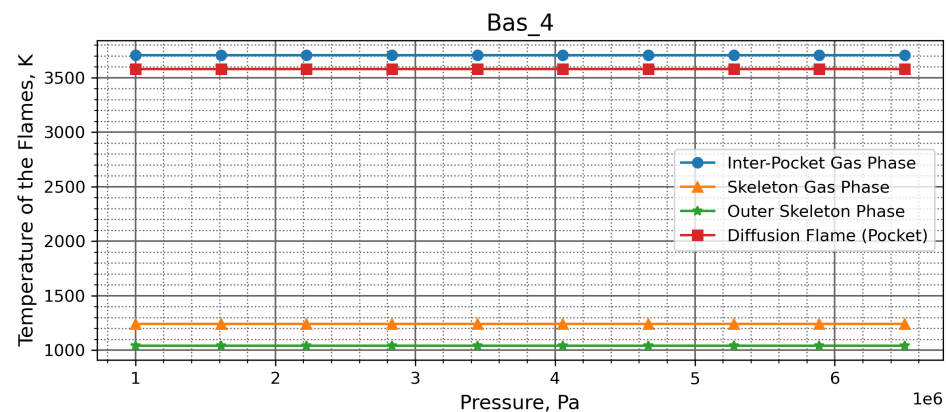
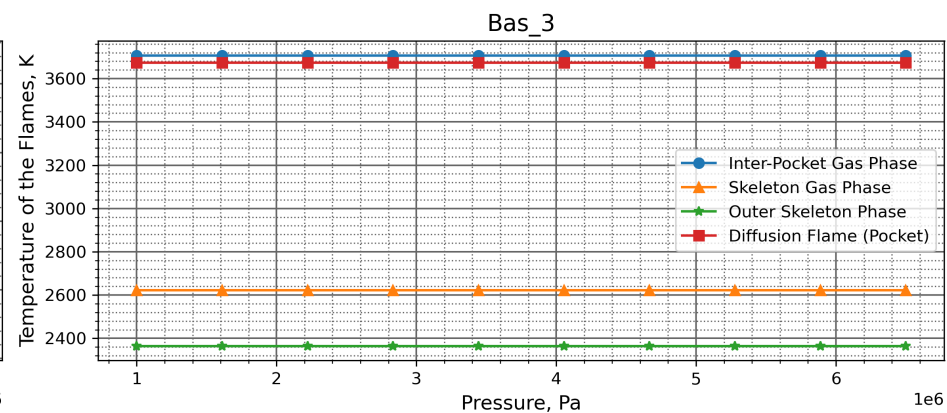
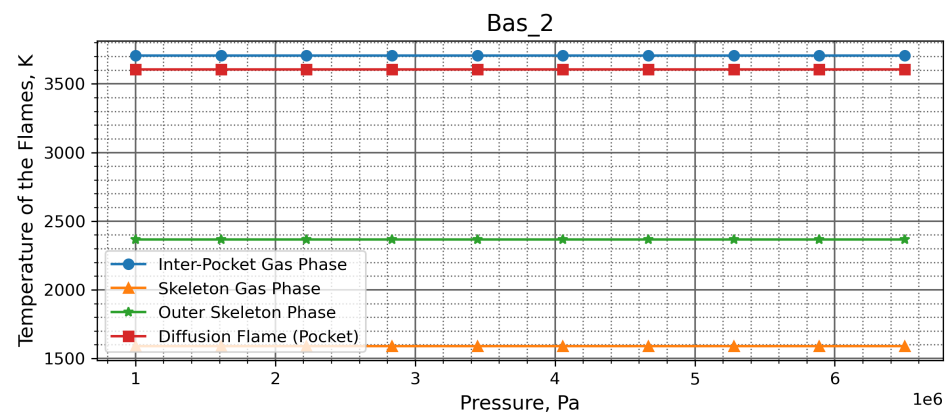
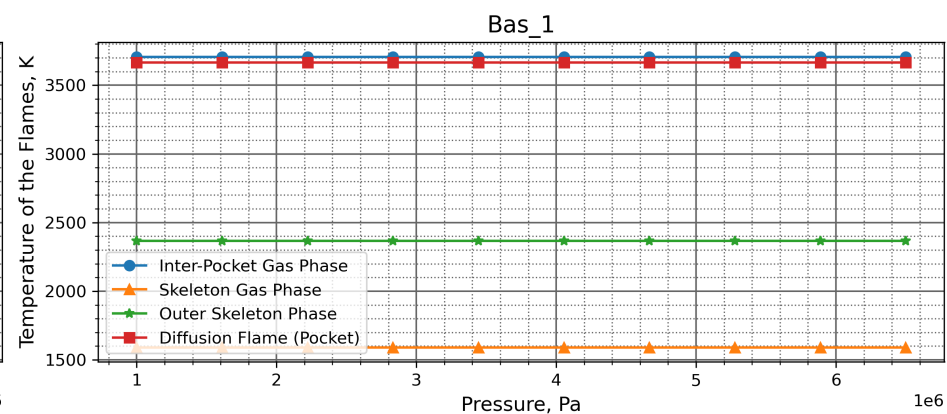
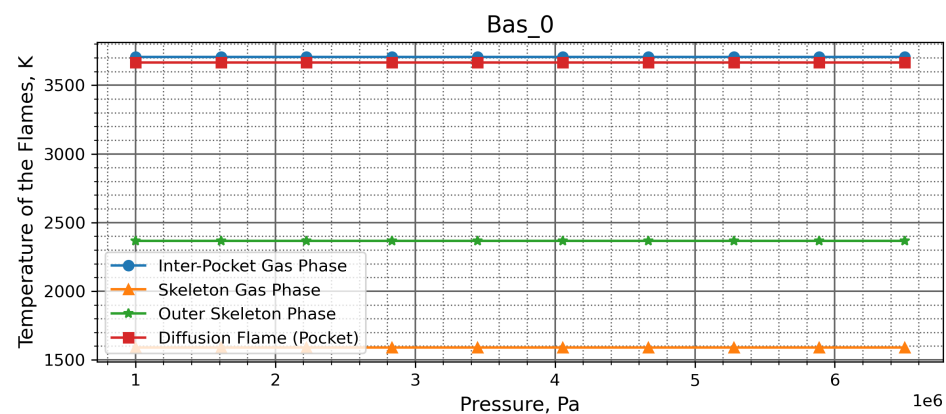
Average Molar Mass, kg/mol



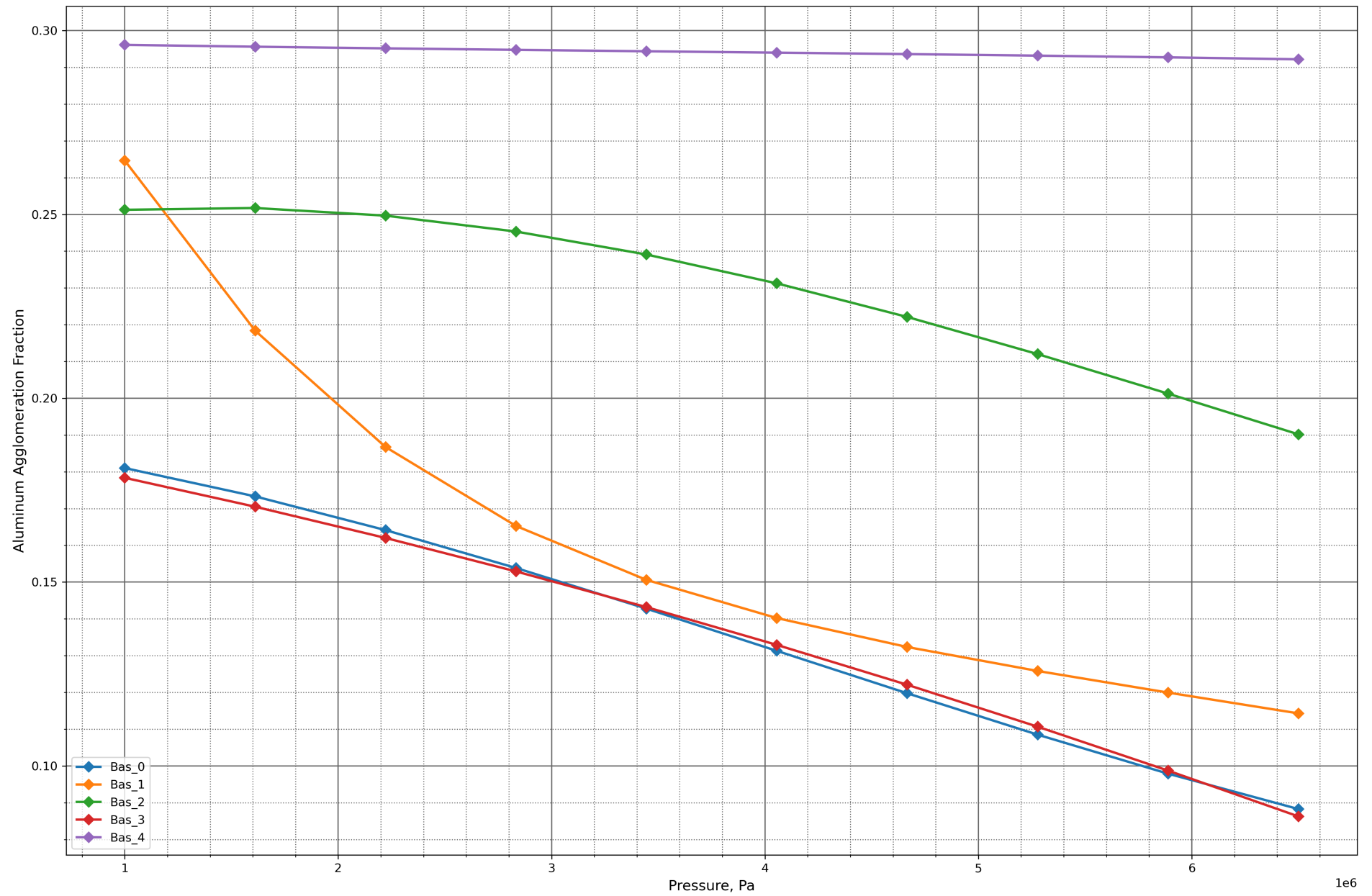
Constant Volume Heat Capacity (c_v), J/(kg·K)



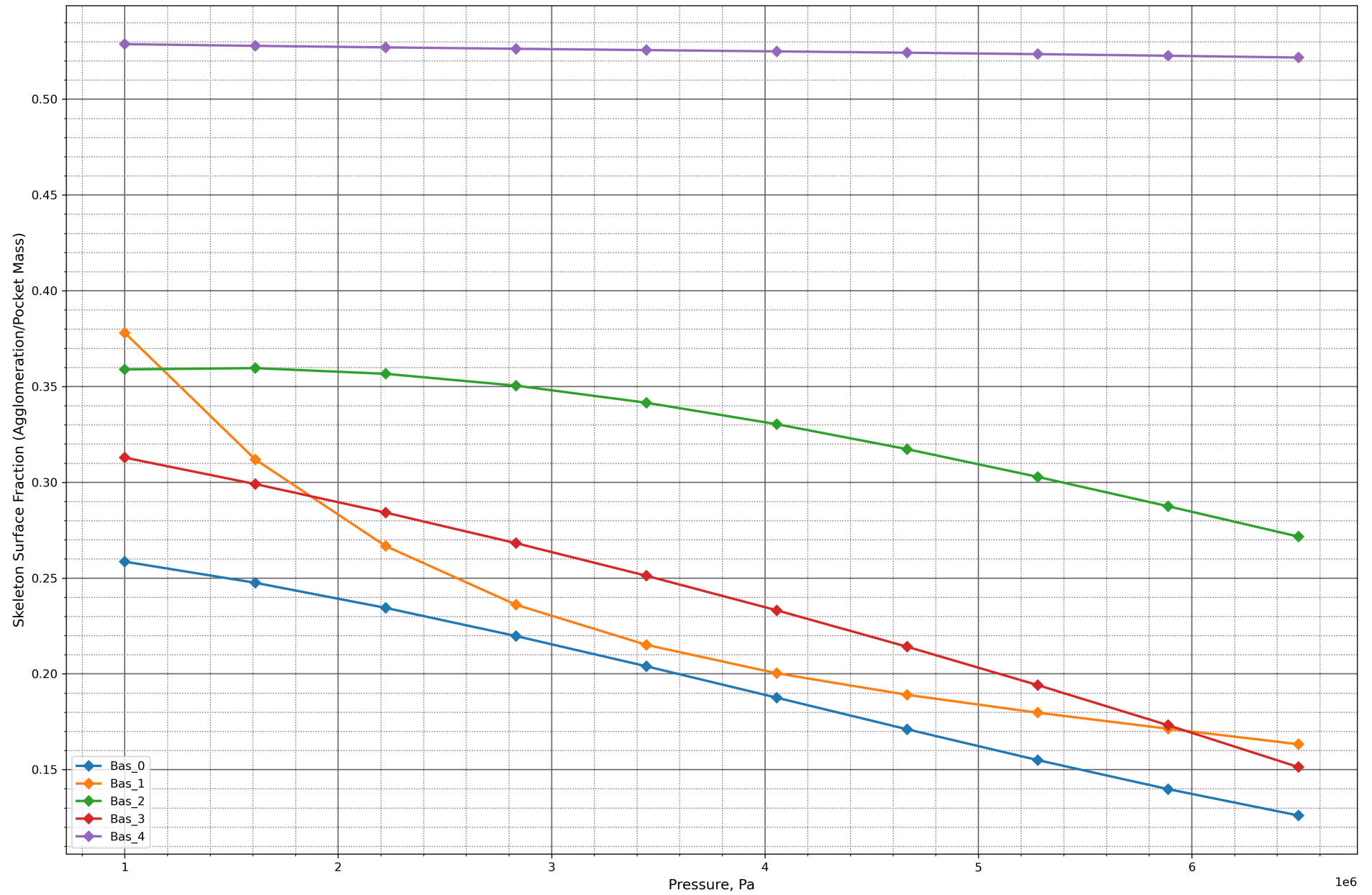
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 09:01:29.925

Garbage Collection Information

Total Memory: 7917552 bytes
GC Pause Time Percentage: 0.16%
Generation 0 Collections: 40869
Generation 1 Collections: 3228
Generation 2 Collections: 323

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 17728476
Maximum execution time: 100.9636 ms
Minimum execution time: 0.0001 ms
Mean execution time: 1.0018983741466823 ms
Execution time standard deviation: 0.1701441685177096 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 17341604
Maximum execution time: 37.7542 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.3454085185487986 ms
Execution time standard deviation: 0.07616811412548496 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 14640238
Maximum execution time: 48.2458 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34255560139121033 ms
Execution time standard deviation: 0.08459136493712968 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 45.0405 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.005581468509778 ms
Execution time standard deviation: 0.16402247149534344 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 17234407
Maximum execution time: 44.4357 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3464874117107452 ms
Execution time standard deviation: 0.07487624250002009 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 14520585
Maximum execution time: 69.7696 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.34371851247038476 ms
Execution time standard deviation: 0.08694292520190189 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 82.0065 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0052027421811287 ms
Execution time standard deviation: 0.1665684018565072 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 17252390
Maximum execution time: 63.3983 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34647699352372385 ms
Execution time standard deviation: 0.07619174334400014 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 14543925
Maximum execution time: 38.2714 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.343798668193143 ms
Execution time standard deviation: 0.08513137758665427 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 53.0517 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0056658700216363 ms
Execution time standard deviation: 0.1652418441459734 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 17242961
Maximum execution time: 51.328 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34708756896802834 ms
Execution time standard deviation: 0.07686975480099607 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 14536547
Maximum execution time: 47.3876 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3442156859396989 ms
Execution time standard deviation: 0.08322637164260657 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 96.0102 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.005156808428287 ms
Execution time standard deviation: 0.16463096180177583 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 17252095
Maximum execution time: 32.3666 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34659772906423697 ms
Execution time standard deviation: 0.07191032077185294 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 14569182

Maximum execution time: 33.6789 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34410857990515037 ms

Execution time standard deviation: 0.08244972860196949 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 53.6329 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0065128865942772 ms

Execution time standard deviation: 0.1659705797027966 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 17250943

Maximum execution time: 31.4846 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34655256538734397 ms

Execution time standard deviation: 0.07097824717849828 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 14566599

Maximum execution time: 37.3752 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.343701459386619 ms

Execution time standard deviation: 0.081843797204675 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 58.0192 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.005914642659279 ms

Execution time standard deviation: 0.16235227023211152 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 17253895

Maximum execution time: 39.2075 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3464141377990623 ms

Execution time standard deviation: 0.07308948035672765 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 14573661

Maximum execution time: 36.3253 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34341492692874886 ms

Execution time standard deviation: 0.0833182871243545 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 43.1281 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0058696747817828 ms

Execution time standard deviation: 0.16354356226478642 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 17253006

Maximum execution time: 39.4957 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.346558369515412 ms
Execution time standard deviation: 0.07587499367159178 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 14554788
Maximum execution time: 61.3787 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34362369855202163 ms
Execution time standard deviation: 0.08712027117393982 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 50.9846 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0041176044678393 ms
Execution time standard deviation: 0.16512514364942638 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 17234027
Maximum execution time: 39.5714 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34619212714478553 ms
Execution time standard deviation: 0.07388457434970508 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 14530896
Maximum execution time: 55.65 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34476497186407645 ms
Execution time standard deviation: 0.0857640645196592 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 54.9967 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0055361927309892 ms
Execution time standard deviation: 0.16774667646768732 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 17233742
Maximum execution time: 51.6027 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3465836964601398 ms
Execution time standard deviation: 0.0768838350554252 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 14532674
Maximum execution time: 55.3792 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34364671717676637 ms
Execution time standard deviation: 0.08612797046411746 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 56.0211 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0031286829831185 ms

Execution time standard deviation: 0.16474675594360735 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 17234752

Maximum execution time: 38.0973 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34750833623251287 ms

Execution time standard deviation: 0.07468569263191385 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 14529361

Maximum execution time: 33.3858 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3430087314507293 ms

Execution time standard deviation: 0.08271377581220372 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 74.0184 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0044830466879338 ms

Execution time standard deviation: 0.16270119959363125 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 17240987

Maximum execution time: 37.3974 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34632747519615686 ms

Execution time standard deviation: 0.07400916934710619 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 14552962

Maximum execution time: 43.573 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.343330471473778 ms

Execution time standard deviation: 0.08092871068735376 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 72.0198 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0057785139172366 ms

Execution time standard deviation: 0.1657751705892831 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 17253962

Maximum execution time: 58.5507 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34643054978331056 ms

Execution time standard deviation: 0.07505537079349651 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 14583983

Maximum execution time: 44.87 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3433336981125279 ms

Execution time standard deviation: 0.08294466854370286 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 55.972 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.005007238742561 ms
Execution time standard deviation: 0.16436812185925082 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 17256922
Maximum execution time: 39.3255 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3466925587772404 ms
Execution time standard deviation: 0.07656712648171475 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 14574794
Maximum execution time: 55.4038 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3438450738925984 ms
Execution time standard deviation: 0.08664859592351366 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 57.0402 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0048600617833323 ms
Execution time standard deviation: 0.16669593756844425 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 17237757
Maximum execution time: 36.3952 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3464526158885172 ms
Execution time standard deviation: 0.0737962414606362 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 14531472
Maximum execution time: 58.3447 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3429257940558018 ms
Execution time standard deviation: 0.08509284555753241 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 57.6146 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0045140885989556 ms
Execution time standard deviation: 0.1635431705578485 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 17250327
Maximum execution time: 39.3289 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34628093588024844 ms
Execution time standard deviation: 0.07663904261626597 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 14563317
Maximum execution time: 30.4154 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.3435370375718637 ms
Execution time standard deviation: 0.08213643198665059 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 45.0562 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0059324979259359 ms
Execution time standard deviation: 0.1666816608102624 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 17238666
Maximum execution time: 86.4016 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34580293171755366 ms
Execution time standard deviation: 0.082081719304286 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 14531932
Maximum execution time: 46.4047 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34317362807639923 ms
Execution time standard deviation: 0.08636967332170513 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 37.1712 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0041190695021396 ms
Execution time standard deviation: 0.163415947106078 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 17231081
Maximum execution time: 40.4064 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3459314576955275 ms
Execution time standard deviation: 0.07426557394537008 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 14508480
Maximum execution time: 41.5493 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34398900913118013 ms
Execution time standard deviation: 0.08360097191520123 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 62.0484 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0052642932669338 ms
Execution time standard deviation: 0.16316457906988122 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 17250308
Maximum execution time: 55.3824 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34694848552846147 ms
Execution time standard deviation: 0.07581086456526652 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 14599558
Maximum execution time: 31.7318 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3436277225447871 ms
Execution time standard deviation: 0.08427407792315365 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 63.8066 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0036755012864262 ms
Execution time standard deviation: 0.16712021111829614 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 17238860
Maximum execution time: 56.7785 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34605792580835765 ms
Execution time standard deviation: 0.07531389478590038 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 14553982
Maximum execution time: 48.7987 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3446481764371711 ms
Execution time standard deviation: 0.08719835090182589 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 49.635 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0064895675801675 ms
Execution time standard deviation: 0.16211504381715916 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 17246546
Maximum execution time: 47.6122 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3463536862453221 ms
Execution time standard deviation: 0.07368687788635368 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 14541506
Maximum execution time: 43.9873 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3434132773799102 ms
Execution time standard deviation: 0.08444415134253189 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 60.0475 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0046083819942917 ms
Execution time standard deviation: 0.16364238371255788 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 17256028
Maximum execution time: 45.3939 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.3472935600881962 ms
Execution time standard deviation: 0.07427671295893847 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 14551189
Maximum execution time: 37.4118 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3435940225365812 ms
Execution time standard deviation: 0.08394688364602018 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 65.0252 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0049573824112235 ms
Execution time standard deviation: 0.16398333312238686 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 17255907
Maximum execution time: 36.1267 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34602887490061135 ms
Execution time standard deviation: 0.0734288418190095 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 14557740
Maximum execution time: 51.3561 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34371632780911693 ms
Execution time standard deviation: 0.08473270102624446 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 56.9452 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0047424180092772 ms
Execution time standard deviation: 0.16892476851195512 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 17243995
Maximum execution time: 46.9527 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34617150464259616 ms
Execution time standard deviation: 0.07866647857331159 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 14563200
Maximum execution time: 40.3619 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.343996665616033 ms
Execution time standard deviation: 0.08448088816752991 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 17628048
Maximum execution time: 39.9775 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0045028467191728 ms
Execution time standard deviation: 0.1644545706846462 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 17243982

Maximum execution time: 50.3941 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34652477508965934 ms

Execution time standard deviation: 0.07585628736841489 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 14551691

Maximum execution time: 36.3782 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34300129721695244 ms

Execution time standard deviation: 0.0843006525625996 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 66.0437 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0048149050764563 ms

Execution time standard deviation: 0.1698393539471332 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 17220713

Maximum execution time: 58.7688 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34628634209859066 ms

Execution time standard deviation: 0.0756419122929188 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 14495799

Maximum execution time: 60.6247 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3429935640042565 ms

Execution time standard deviation: 0.08386055846501032 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 58.2843 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.004718807629555 ms

Execution time standard deviation: 0.16540786254368178 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 17243246

Maximum execution time: 49.3872 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34681615125714155 ms

Execution time standard deviation: 0.0760961134111778 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 14565585

Maximum execution time: 53.3838 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3434849516446182 ms

Execution time standard deviation: 0.08616370841828908 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 51.0393 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.005398411605217 ms
Execution time standard deviation: 0.16386718161316297 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 17243155
Maximum execution time: 44.0513 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34602356018370095 ms
Execution time standard deviation: 0.07535586148070923 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 14569709
Maximum execution time: 47.398 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34402236783862533 ms
Execution time standard deviation: 0.08220789401866979 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 40.806 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0045787042842667 ms
Execution time standard deviation: 0.1617103360448424 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 17251259
Maximum execution time: 62.9253 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34626179230747467 ms
Execution time standard deviation: 0.0749320374363002 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 14553570
Maximum execution time: 38.118 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3435035701618212 ms
Execution time standard deviation: 0.08142678732192787 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 60.3158 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0050917427611483 ms
Execution time standard deviation: 0.16908835482679252 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 17239761
Maximum execution time: 38.5215 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34596964736930597 ms
Execution time standard deviation: 0.07617941418778841 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 14548094
Maximum execution time: 111.3512 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3433715036210095 ms

Execution time standard deviation: 0.08884014938239342 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 84.9882 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0051605984791718 ms

Execution time standard deviation: 0.1646651943944277 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 17229434

Maximum execution time: 57.3937 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34618020036520974 ms

Execution time standard deviation: 0.0747334244908243 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 14509933

Maximum execution time: 113.4529 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34471538907868604 ms

Execution time standard deviation: 0.09072212393575688 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 45.0615 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.005669445210298 ms

Execution time standard deviation: 0.1677106848549426 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 17240458

Maximum execution time: 42.4473 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34625176474427666 ms

Execution time standard deviation: 0.0763246730874297 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 14527377

Maximum execution time: 44.1291 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34385500272346364 ms

Execution time standard deviation: 0.08775251893126458 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 17628048

Maximum execution time: 56.9779 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.005925890115603 ms

Execution time standard deviation: 0.16435014645358423 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 17249889

Maximum execution time: 104.4707 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3475697018456641 ms

Execution time standard deviation: 0.07795186309053476 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 14555163
Maximum execution time: 38.3998 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34333164808940914 ms
Execution time standard deviation: 0.08464593382259622 ms

Method: Telemetry/Proc/33/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 66.0492 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0044779055171498 ms
Execution time standard deviation: 0.16282910654890648 ms

Method: Telemetry/Proc/33/OptimizationProblemExecutionPerformance

Call count: 17249141
Maximum execution time: 49.399 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3464639507788126 ms
Execution time standard deviation: 0.07625035250264771 ms

Method: Telemetry/Proc/33/OptimizationProblemExecutionPerformance

Call count: 14546721
Maximum execution time: 30.4871 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.343467340901072 ms
Execution time standard deviation: 0.0810529956810261 ms

Method: Telemetry/Proc/34/OptimizationProblemExecutionPerformance

Call count: 17628048
Maximum execution time: 48.755 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0074824718766864 ms
Execution time standard deviation: 0.16686684709924665 ms

Method: Telemetry/Proc/34/OptimizationProblemExecutionPerformance

Call count: 17252418
Maximum execution time: 37.6495 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34642393811701716 ms
Execution time standard deviation: 0.07771338770141507 ms

Method: Telemetry/Proc/34/OptimizationProblemExecutionPerformance

Call count: 14548174
Maximum execution time: 55.2436 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3439956940781518 ms
Execution time standard deviation: 0.08549498584768919 ms